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Surpassing the fractional derivative: Concept of the memory-dependent derivative

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ABSTRACT

Enlightened by the Caputo type of fractional derivative, here we bring forth a concept of "memory-dependent derivative", which is simply defined in an integral form of a common derivative with a kernel function on a slipping interval. In case the time delay tends to zero it tends to the common derivative. High order derivatives also accord with the first order one. Comparatively, the form of kernel function for the fractional type is fixed, yet that of the memory-dependent type can be chosen freely according to the necessity of applications. So this kind of definition is better than the fractional one for reflecting the memory effect (instantaneous change rate depends on the past state). Its definition is more intuitionistic for understanding the physical meaning and the corresponding memory-dependent differential equation has more expressive force.

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1. Introduction

Just as reviewed by Diethelm in [1], the concept of the fractional derivative can be dated back to 1695 when de L'Hospital asked the famous question: "What does the derivative $\mathrm{d}^n f/\mathrm{d}x^n$ for $f(x)$ mean if $n = 1/2$?". From that time, a branch of mathematics named *fractional calculus* has been developed. It is a generalization of the commonly used integer-order differentiation and integration. The basic ideology is to look at a fractional derivative as the inverse operation of a fractional integral, which is usually chosen as the Riemann–Liouville form [1]:

$$J_a^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_a^t (t-s)^{\alpha-1} f(s)\mathrm{d}s, \quad t \in [a, b],\ \alpha > 0, \tag{1}$$

where $f(t)$ is integrable on the interval $[a, b]$ and Γ is the Euler's Gamma function. Correspondingly, the Riemann–Liouville fractional derivative is defined as:

$$D_a^\alpha f(t) = D^m J_a^{m-\alpha} f(t) = \frac{\mathrm{d}^m}{\mathrm{d}t^m} \left[\int_a^t \frac{(t-s)^{m-\alpha-1}}{\Gamma(m-\alpha)} f(s)\mathrm{d}s \right], \tag{2}$$

where m is an integer which satisfies $m-1 < \alpha \le m$, D^m is the common m-order derivative. This concept is historically the first and the one for which the mathematical theory has been established quite well by now, but it has certain features that lead to difficulties when applying it to "realworld" problems. As a consequence, the Caputo derivative was developed [1]:

$$D_a^\alpha f(t) = J_a^{m-\alpha} D^m f(t) = \int_a^t K_\alpha(t-s) f^{(m)}(s)\mathrm{d}s \tag{3}$$

with

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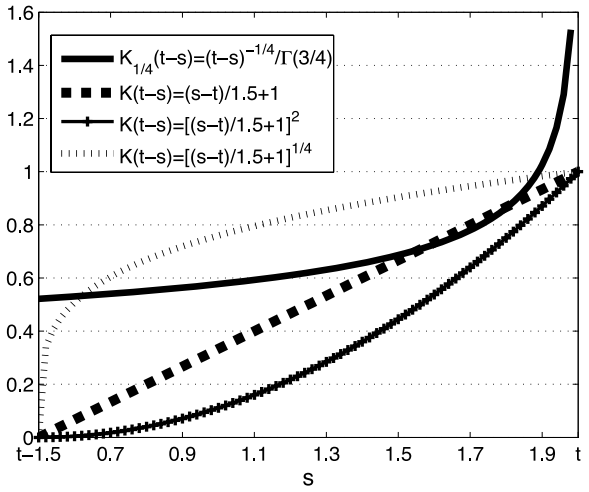


Fig. 1. Comparisons between the kernel $K_\alpha(t-s)$ for fractional derivative and different kernels $K(t-s)$ for memory-dependent derivative, where α and τ are chosen as 1/4 and 1.5, respectively.

$$K_\alpha(t-s) = \frac{(t-s)^{m-\alpha-1}}{\Gamma(m-\alpha)}, \tag{4}$$

where $f^{(m)}$ denotes the common m-order derivative, which has specific physical meaning.

From Eq. (3) we see the α-order fractional derivative at time t is not defined locally, it relies on the total effects of the commonly used m-order integer derivative on the interval $[a, t]$. So it can be used to describe the variation of a system in which the instantaneous change rate depends on the past state, which is called the "memory effect" in a visualized manner [2]. In addition, as indicated in [3] the fractional theory has been applied to almost all fields of science, which include viscoelasticity and rheology, electrical engineering, electrochemistry, biology, biophysics and bioengineering, signal and image processing, mechanics, mechatronics, physics, and control theory.

It follows from (3) and (4) that for a given real number α, the kernel $K_\alpha(t-s)$ is a fixed function. But from the viewpoint of applications, different processes need different kernels to reflect their memory effects, so the kernel should be chosen freely. In addition, since the fractional derivative is defined on the interval $[a, t]$ with a fixed real number a, it may be invalid in describing the memory effect for large time t. In fact, the memory effect of a real process basically occurs in a segment of time, that is, the delayed interval $[t-\tau, t]$ ($\tau > 0$ denotes the time delay). Enlightened by these, we enlarge the concept of derivative by bringing forth a new version and name it "memory-dependent derivative" to reflect the memory effect in a distinct way.

2. Memory-dependent derivative

Notice that the deduced Caputo derivative is in an integral form of the common derivative, we can define it in a direct way:

Definition 2.1. For a differentiable function $f(t)$,

$$D_\tau f(t) = \frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)f'(s)\mathrm{d}s \tag{5}$$

is called the first order "memory-dependent derivative" of f at t relative to the time delay $\tau > 0$, where the kernel function $K(t-s)$ is differentiable about t and s.

This kind of definition can reflect the memory effect on the delayed interval $[t-\tau, t]$ which varies along with time. It has abandoned the dependence relation of fixed point a for a fractional derivative. As discussed in the previous section, for a given α the kernel form $K_\alpha(t-s)$ is fixed, yet here besides the modulatory parameter τ the kernel form $K(t-s)$ can also be chosen freely, such as $1, s-t+1, [(s-t)/\tau+1]^2$ and $[(s-t)/\tau+1]^{1/4}$, etc. The kernel function can be understood as the degree of the past effect on the present, so the forms $[(s-t)/\tau+1]^2$ and $[(s-t)/\tau+1]^{1/4}$ may be more practical, after all, they are a monotone function with $K=0$ for the past time $t-\tau$ and $K=1$ for the present time t. The comparison between $K_\alpha(t-s)$ and $K(t-s)$ are shown in Fig. 1.

The right hand side of (5) can be understood as a mean value of $f'(s)$ on the past interval $[t-\tau, t]$ with different weights. Generally, from the viewpoint of applications, the memory effect requires weight $0 \le K(t-s) < 1$ for $s \in [t-\tau, t)$, so the

magnitude of the memory-dependent derivative $D_\tau f(t)$ is usually smaller than that of the common derivative $f'(t)$. By and large, the relation between them can be approximated by the mean-value function:

$$M(t) = \frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)\mathrm{d}s, \tag{6}$$

which usually takes a constant value. In addition, in case $K \equiv 1$,

$$D_\tau f(t) = \frac{1}{\tau}\int_{t-\tau}^{t} f'(s)\mathrm{d}s = \frac{f(t)-f(t-\tau)}{\tau} \to f'(t). \tag{7}$$

So the common derivative $d/\mathrm{d}t$ can be seen as the limit of D_τ as $\tau \to 0$.

Certainly, the linear property for the memory-dependent derivative is also satisfied. In fact,

$$D_\tau[af(t) + bg(t)] = aD_\tau f(t) + bD_\tau g(t)$$

is easily checked for arbitrary real numbers a, b and differentiable functions $f(t)$, $g(t)$.

Definition 2.2. Let m be a natural number ($m \in N$), then for an m-times differentiable function $f(t)$,

$$D_\tau^m f(t) = \frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)f^{(m)}(s)\mathrm{d}s \tag{8}$$

is called the m-order "memory-dependent derivative" of f at t relative to the time delay $\tau > 0$, where $K(t-s)$ is m-times differentiable about t and s.

To repeat the calculation process in (7) for the case $K \equiv 1$, also yields $D_\tau^m f(t) \to f^{(m)}(t)$ as $\tau \to 0$. So the memory-dependent derivative also accords with the common derivative for high order cases.

Theorem 2.1. *The* 1*-order and* 2*-order memory-dependent derivatives satisfy the relation below:*

$$D_\tau^2 = DD_\tau. \tag{9}$$

Proof. It follows from Definition 2.1 that

$$\begin{aligned}
DD_\tau f(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)f'(s)\mathrm{d}s\right] \\
&= \frac{1}{\tau}\left[\int_{t-\tau}^{t}\frac{\partial K}{\partial t}f'(s)\mathrm{d}s + K(t-s)f'(s)|_{s=t}\cdot 1 - K(t-s)f'(s)|_{s=t-\tau}\cdot 1\right] \\
&= \frac{1}{\tau}\left[-\int_{t-\tau}^{t}\frac{\partial K}{\partial s}f'(s)\mathrm{d}s + K(t-s)f'(s)\,|_{t-\tau}^{t}\right] \\
&= \frac{1}{\tau}\left[-K(t-s)f'(s)\,|_{t-\tau}^{t} + \int_{t-\tau}^{t} K(t-s)f''(s)\mathrm{d}s + K(t-s)f'(s)\,|_{t-\tau}^{t}\right] \\
&= \frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)f''(s)\mathrm{d}s = D_\tau^2 f(t).
\end{aligned} \tag{10}$$

The proof is finished. □

To repeat the above deduction process we get a corollary below:

Corollary 2.1. *The* m*-order and* 1*-order memory-dependent derivatives satisfy the relation below:*

$$D_\tau^m = \overbrace{D\cdots D}^{m-1}D_\tau, \quad m \in N. \tag{11}$$

3. Memory-dependent differential equations

Now that the memory-dependent derivative is defined, it is a natural thing to apply it in differential equations. It is very interesting to find that a differential equation with a memory-dependent type of derivative (simple called "memory-dependent differential equation") can represent many kinds of equations. Take the simplest Malthus population model [4] as an example:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = ru(t), \tag{12}$$

where r is the intrinsic growth rate (birth rate–death rate). We can take it as a particular case of the following memory-dependent differential equation for $K \equiv 1$ and $\tau \to 0$:

$$D_\tau u(t) = ru(t). \tag{13}$$

In fact, for some species, especially the long age ones, the memory effect cannot be neglected. The reason is that, the intrinsic growth rate r is an estimate of a long time mean, it does not merely associate with a moment t. So the changed rate of the population should be also a mean value on the past time segment $[t-\tau, t]$, where the time delay τ reflects how far the effect is from the present. From the viewpoint of mathematics, Eq. (13) can represent many kinds of equations, such as:

(1) For the constant case with $K \equiv 1$,

$$D_\tau u(t) = \frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)u'(s)\mathrm{d}s = \frac{u(t)-u(t-\tau)}{\tau},$$

at this time Eq. (13) is identical to the difference equation with $r\tau \neq 1$:

$$u(t) = \frac{1}{1-r\tau}u(t-\tau). \tag{14}$$

(2) For the linear case with $K(t-s) = (s-t)/\tau + 1$, it is easy to see that $K(t, t-\tau) = 0$ and simple deduction yields ($r\tau \neq 1$):

$$u'(t) = \frac{1}{\tau(1-r\tau)}[u(t)-u(t-\tau)]. \tag{15}$$

It is a first order differential equation with time delay.

(3) For the parabolic curve case with $K(t-s) = [(s-t)/\tau + 1]^2$, Eq. (13) can be transformed to the following second order differential equation with time delay ($r\tau \neq 1$):

$$u''(t) = \frac{2}{\tau(1-r\tau)}u'(t) - \frac{2}{\tau^2(1-r\tau)}[u(t)-u(t-\tau)]. \tag{16}$$

Due to the freedom of choice for the kernel function, the simple equation (13) can also represent other complex equations. When the memory-dependent Malthus model (13) is substituted by the Logistic one [4]:

$$D_\tau u(t) = ru(t)[1-u(t)], \tag{17}$$

where a self-limiting effect is incorporated to reflect the carrying capacity of the environment, we can get the delayed model below:

$$u'(t) = \frac{u(t)-u(t-\tau)}{\tau(1-r\tau)+2\tau^2 ru(t)} \tag{18}$$

for the linear kernel case with $K(t-s) = (s-t)/\tau + 1$. Besides the kernel function, τ is also a modulatory parameter. Effect of time delay on the solution of Eq. (17) (that is, Eq. (18)) is simulated in Fig. 2.

In addition, high order memory-dependent differential equations may include more types of equations. Such as the second order case:

$$D_\tau^2 u(t) = f(t, u(t)), \tag{19}$$

if the kernel function is chosen as $K \equiv 1$, then it reads:

$$u'(t) - u'(t-\tau) = \tau f(t, u(t)). \tag{20}$$

It is a neutral type equation [5].

4. Comparison with the fractional type equation

We compare the memory-dependent type and the fractional type equation by considering the following linear initial-value problems:

$$\begin{cases} D_\tau u(t) = ru(t), & t > 0, \\ u(s) = \mathrm{e}^{rs}, & s \in [-\tau, 0] \end{cases} \tag{21}$$

and

$$\begin{cases} D_0^\alpha v(t) = rv(t), & t > 0, \\ v(0) = 1, & 0 < \alpha < 1. \end{cases} \tag{22}$$

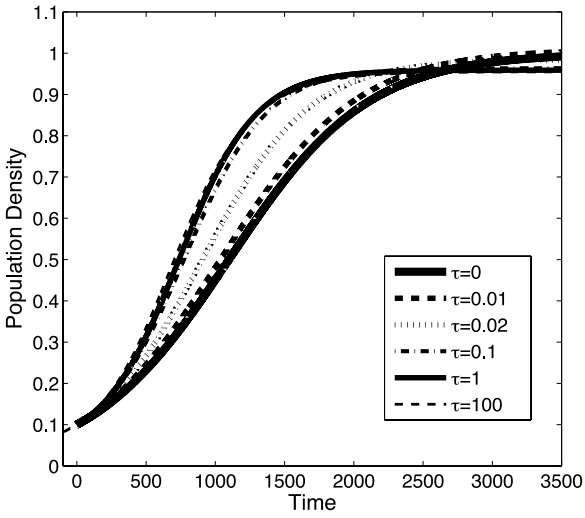


Fig. 2. Effect of time delay on the memory-dependent Logistic model (17) for the linear kernel case with $K(t-s) = (s-t)/\tau + 1$ (that is, Eq. (18)) with initial condition $u(s) = 0.1e^{rs}$ on $[-\tau, 0]$, where $r = 0.02$. Here $\tau = 0$ indicates the common case.

Here the derivatives are

$$D_\tau u(t) = \frac{1}{\tau}\int_{t-\tau}^{t} K(t-s)u'(s)\mathrm{d}s = \int_{t-\tau}^{t}\left[\frac{s-t+\tau}{\tau^2}\right]u'(s)\mathrm{d}s, \tag{23}$$

$$D_0^\alpha v(t) = \int_0^t K_\alpha(t-s)v'(s)\mathrm{d}s = \int_0^t\left[\frac{(t-s)^{-\alpha}}{\Gamma(1-\alpha)}\right]v'(s)\mathrm{d}s. \tag{24}$$

At this time, the equation in (21) equals to that of (15). In addition, it follows from Theorem 4.3 in [1] that the solution of (22) can be explicitly expressed by the Mittag-Leffler function:

$$v(t) = E_\alpha(rt^\alpha) = \sum_{k=0}^{\infty}\frac{(rt^\alpha)^k}{\Gamma(\alpha k+1)}. \tag{25}$$

To make a comparison, we approximate the summation (25) by its first 50 terms. The numerical simulations in Fig. 3 show that the solution of the memory-dependent type increases more rapidly than the common type and, on the contrary, the fractional type increases much slower. In fact, the occurrence of this phenomenon is very natural. By using the Mean-Value Theorem for integrals we have

$$\int_{t-\tau}^{t}\frac{s-t+\tau}{\tau^2}u'(s)\mathrm{d}s = u'(\xi)\int_{t-\tau}^{t}\frac{s-t+\tau}{\tau^2}\mathrm{d}s = \frac{1}{2}u'(\xi), \tag{26}$$

$$\int_0^t\frac{(t-s)^{-\alpha}}{\Gamma(1-\alpha)}v'(s)\mathrm{d}s = \frac{v'(\eta)}{\Gamma(1-\alpha)}\int_0^t(t-s)^{-\alpha}\mathrm{d}s = \frac{tv'(\eta)}{(1-\alpha)\Gamma(1-\alpha)} \tag{27}$$

for $t-\tau \le \xi \le t$ and $0 \le \eta \le t$. So the magnitude of $D_\tau u(t)$ is almost $1/2$ that of $u'(t)$ and that of $D_0^\alpha u(t)$ is almost $t/(1-\alpha)\Gamma(1-\alpha)$ times that of $u'(t)$. *Hence, larger t implies larger magnitude of the fractional derivative $D_0^\alpha u(t)$. As a consequence, the definition of fractional derivative is unsuitable for describing a long-time process. This drawback is essential. But for the memory-dependent type, the kernel function is defined on a slipping interval $[t-\tau, t]$ and its integral value is fixed. In this sense the definition of the memory-dependent derivative is more rational.*

In addition, Fig. 3 also indicates that the solution is very sensitive to the fractional order α near 1, after all, a difference exists between $\alpha = 0.9$ and $\alpha = 1$.

5. Conclusions

Relative to the fractional derivative, here we have brought forth a concept of "memory-dependent derivative", which is directly defined by the integral of a common derivative. The fractional derivative meets the consistency with the common derivative by $D_a^\alpha \to \mathrm{d}/\mathrm{d}t$ as $\alpha \to 1$, yet here by $D_\tau \to \mathrm{d}/\mathrm{d}t$ as $\tau \to 0$ with $K \equiv 1$, where τ is the time delay reflects the past effect on the present. The form of the kernel function for the fractional type is fixed, yet that of the memory-dependent type can be chosen freely according to the necessity of applications. A simple memory-dependent differential equation can represent many kinds of equations with time delay, such as the difference type, the first-order type, the second-order

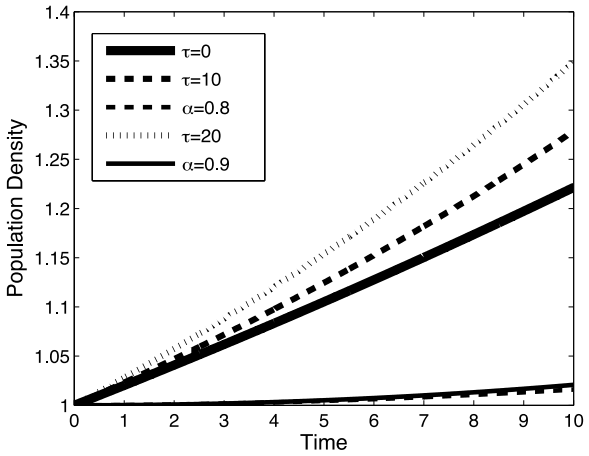


Fig. 3. Comparison between the memory-dependent type and the fractional type of initial-value problems (21) and (22), where $r = 0.02$. Here $\tau = 0$ indicates the common case.

type, etc. It follows from the numerical comparison with the fractional type that the memory-dependent type has more power in describing the memory effect. The analytical results show that larger t leads to a larger magnitude of the fractional derivative D_0^α. This is an essential drawback for it. Yet this phenomenon does not occur for the memory-dependent type. In all, the memory-dependent type of derivative is better than the fractional type in describing the memory effect, its definition is more intuitionistic for understanding the physical meaning and the corresponding memory-dependent differential equation has more expressive force.

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